CLAIMS

WHAT IS CLAIMED IS:

1. A computer system for controlling an aerial vehicle to travel in an air corridor while maintaining safe separation from other agent vehicles, the computer system comprising:

a processor; and

a non-transitory computer readable medium having stored thereon program code for transferring data to another computer system, the program code causing the processor to:

receive a trained air corridor model associated with an origin, a destination, and an air corridor representative of airspace between the origin and the destination, wherein the trained air corridor model is configured to maintain safe separation distance between vehicles in the air corridor;

identify current position data representing a current position of the aerial vehicle in the air corridor;

receive agent position data associated with a position of at least one agent vehicle in the air corridor; and

determine a next position of the aerial vehicle in the air corridor based on the trained air corridor model, the current position data, and the agent position data.

2. The computer system of claim 1, wherein the air corridor includes a joint set of spatio-temporal data cubes that represent portions of airspace in the air corridor from the origin to the destination; and

wherein at least one data cube of the joint set of spatio-temporal data cubes is subdivided into a plurality of data sub-cubes, wherein a size of the data sub-cubes is based on a defined minimum safe separation distance.

3. The computer system of claim 2, wherein identifying the current position data of the current position of the aerial vehicle in the air corridor includes identifying a current data sub-cube of the air corridor in which the aerial vehicle is currently positioned; and

wherein determining the next position of the aerial vehicle in the air corridor based on the trained air corridor model, the current position data, and the agent position data includes determining a next data sub-cube of the air corridor that is adjacent to the identified current data sub-cube.

4. The computer system of claim 2, wherein determining the next position of the aerial vehicle in the air corridor based on the trained air corridor model includes adapting the air corridor based on the trained air corridor model to optimize use of the air corridor, wherein the adapting includes at least one of the following: subdividing a data cube of the air corridor into octant sub-cubes of the air corridor and combining sub-cubes of the air corridor into a larger cube of the air corridor.

5. The computer system of claim 1, wherein the program code further causes the processor to send the identified current position data to the at least one agent vehicle in the air corridor.

6. The computer system of claim 1, further comprising, controlling the aerial vehicle to travel from the current position to the determined next position in the air corridor including at least one of the following: accelerating the aerial vehicle, decelerating the aerial vehicle, causing the aerial vehicle to gain altitude, causing the aerial vehicle to lose altitude, and causing the aerial vehicle to change direction.

7. The computer system of claim 1, wherein the air corridor includes airspace from an altitude of 400 feet to an altitude of 2500 feet.

8. A computerized method for controlling an aerial vehicle to travel in an air corridor while maintaining safe separation from other agent vehicles, the method comprising:

receiving, by a processor of an aerial vehicle, a trained air corridor model associated with an origin, a destination, and an air corridor representative of airspace between the origin and the destination, wherein the trained air corridor model is configured to maintain safe separation distance between vehicles in the air corridor;

identifying, by the processor, current position data representing a current position of the aerial vehicle in the air corridor;

receiving, by the processor, agent position data associated with a position of at least one agent vehicle in the air corridor; and

determining, by the processor, a next position of the aerial vehicle in the air corridor based on the trained air corridor model, the current position data, and the agent position data.

9. The computerized method of claim 8, wherein the air corridor includes a joint set of spatio-temporal data cubes that represent portions of airspace in the air corridor from the origin to the destination; and

wherein at least one data cube of the joint set of spatio-temporal data cubes is subdivided into a plurality of data sub-cubes, wherein a size of the data sub-cubes is based on a defined minimum safe separation distance.

10. The computerized method of claim 9, wherein identifying the current position data of the current position of the aerial vehicle in the air corridor includes identifying a current data sub-cube of the air corridor in which the aerial vehicle is currently positioned; and

wherein determining the next position of the aerial vehicle in the air corridor based on the trained air corridor model, the current position data, and the agent position data includes determining a next data sub-cube of the air corridor that is adjacent to the identified current data sub-cube.

11. The computerized method of claim 9, wherein determining the next position of the aerial vehicle in the air corridor based on the trained air corridor model includes adapting the air corridor based on the trained air corridor model to optimize use of the air corridor, wherein the adapting includes at least one of the following: subdividing a data cube of the air corridor into octant sub-cubes of the air corridor and combining sub-cubes of the air corridor into a larger cube of the air corridor.

12. The computerized method of claim 8, further comprising sending, by the processor, the identified current position data to the at least one agent vehicle in the air corridor.

13. The computerized method of claim 8, further comprising, controlling the aerial vehicle to travel from the current position to the determined next position in the air corridor including at least one of the following: accelerating the aerial vehicle, decelerating the aerial vehicle, causing the aerial vehicle to gain altitude, causing the aerial vehicle to lose altitude, and causing the aerial vehicle to change direction.

14. The computerized method of claim 8, wherein the air corridor includes airspace from an altitude of 400 feet to an altitude of 2500 feet.

15. A non-transitory computer storage medium having stored thereon program code executable by a first computer system at a first site, the program code embodying a method comprising:

receiving, by an aerial vehicle, a trained air corridor model associated with an origin, a destination, and an air corridor that is representative of airspace between the origin and the destination, wherein the trained air corridor model is configured to maintain safe separation between vehicles in the air corridor;

identifying current position data representing a current position of the aerial vehicle in the air corridor;

receiving agent position data associated with a position of at least one agent vehicle in the air corridor; and

determining a next position of the aerial vehicle in the air corridor based on the trained air corridor model, the current position data, and the agent position data.

16. The non-transitory computer storage medium of claim 15, wherein the air corridor includes a joint set of spatio-temporal data cubes that represent portions of airspace in the air corridor from the origin to the destination; and

wherein at least one data cube of the joint set of spatio-temporal data cubes is subdivided into a plurality of data sub-cubes, wherein a size of the data sub-cubes is based on a defined minimum safe separation distance.

17. The non-transitory computer storage medium of claim 16, wherein identifying the current position data of the current position of the aerial vehicle in the air corridor includes identifying a current data sub-cube of the air corridor in which the aerial vehicle is currently positioned; and

wherein determining the next position of the aerial vehicle in the air corridor based on the trained air corridor model, the current position data, and the agent position data includes determining a next data sub-cube of the air corridor that is adjacent to the identified current data sub-cube.

18. The non-transitory computer storage medium of claim 16, wherein determining the next position of the aerial vehicle in the air corridor based on the trained air corridor model includes adapting the air corridor based on the trained air corridor model to optimize use of the air corridor, wherein the adapting includes at least one of the following: subdividing a

data cube of the air corridor into octant sub-cubes of the air corridor and combining sub-cubes of the air corridor into a larger cube of the air corridor.

19. The non-transitory computer storage medium of claim 15, the method embodied by the program code further comprising sending the identified current position data to the at least one agent vehicle in the air corridor.

20. The non-transitory computer storage medium of claim 15, wherein the method embodied by the program code further comprises, controlling the aerial vehicle to travel from the current position to the determined next position in the air corridor including at least one of the following: accelerating the aerial vehicle, decelerating the aerial vehicle, causing the aerial vehicle to gain altitude, causing the aerial vehicle to lose altitude, and causing the aerial vehicle to change direction.